



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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September 1st 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data was collected via the SpaceX REST API and by web scraping historical Falcon 9 launch records from Wikipedia, then cleaned and labeled for landing outcome
- Exploratory analysis (visualization + SQL) and interactive analytics (Folium map, Plotly Dash dashboard) were used to uncover patterns between payload, orbit, launch site and landing success
- Four classification models (Logistic Regression, SVM, Decision Tree, KNN) were trained and tuned with GridSearchCV; the Decision Tree Classifier achieved the best cross-validated accuracy (87.5%)
- Findings show landing success has improved substantially over time and varies meaningfully by launch site, orbit type and payload mass
- GitHub link: [leonorrebola9/datascience_capstone](https://github.com/leonorrebola9/datascience_capstone)

Introduction

- SpaceX advertises Falcon 9 launches at \$62 million, versus \$165+ million from other providers, largely because SpaceX reuses the rocket's first stage
- If we can predict whether the first stage will successfully land, we can estimate the cost of a launch — valuable for a competitor bidding against SpaceX
- This project aims to answer: what factors most influence a successful first-stage landing, and can we build a reliable model to predict it?

Section 1

Methodology

Methodology

Executive Summary

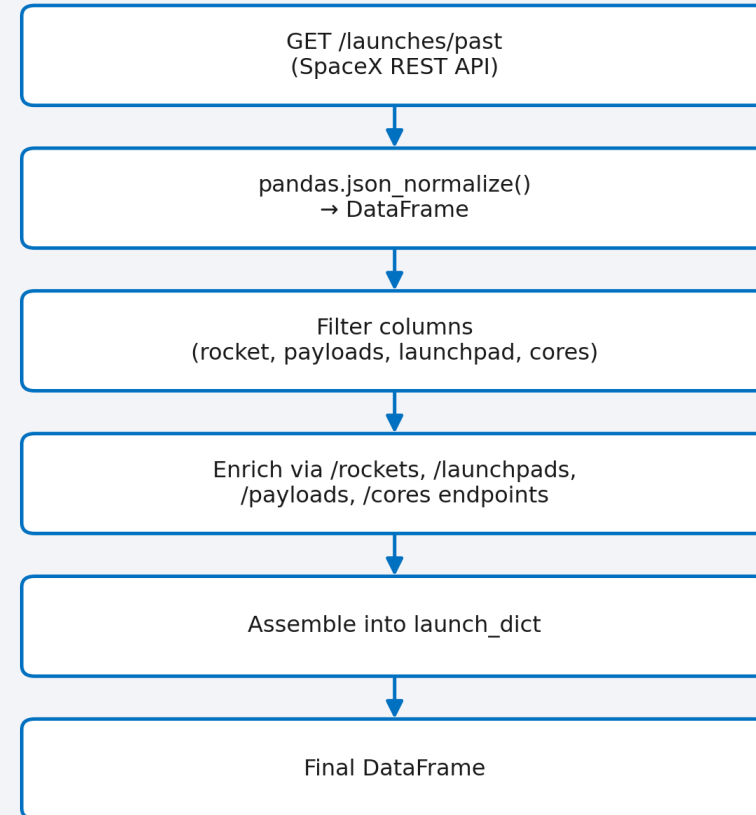
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Two data sources were combined: the SpaceX REST API (structured launch/rocket/payload/core data) and Wikipedia (historical Falcon 9 and Falcon Heavy launch records via web scraping)
- API data was requested with the Python requests library and normalized into a DataFrame with `pandas.json_normalize()`
- Wikipedia data was parsed with BeautifulSoup, extracting the launch records table and cleaning inconsistent HTML annotations

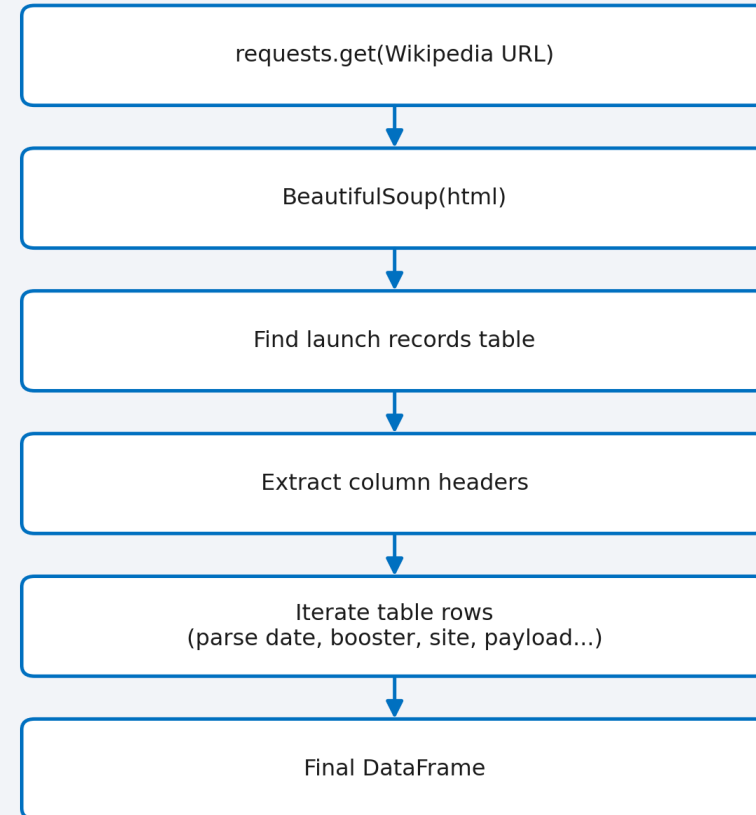
Data Collection – SpaceX API

- Process: GET request to the SpaceX API, then enrich and assemble into a DataFrame (see diagram)
- https://github.com/leonorrebola9/datascience_capstone/blob/main/jupyter-labs-spacex-data-collection-api-v2.ipynb



Data Collection - Scraping

- Process: request the Wikipedia page, parse the HTML table with BeautifulSoup, and build a DataFrame (see diagram)
- https://github.com/leonorrebola9/datasience_capstone/blob/main/jupyter-labs-webscraping.ipynb



Data Wrangling

- Missing values were identified (`df.isnull().sum()/len(df)*100`); PayloadMass NaNs were replaced with the column mean, while LandingPad NaNs were kept as None (meaning no landing pad was used)
- Launch counts were summarized per site (CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13) and per orbit type (GTO, ISS, VLEO, PO, LEO, SSO, MEO, HEO, ES-L1, SO, GEO)
- The Outcome column (e.g. 'True ASDS', 'False Ocean') was converted into a binary Class label: 0 for an unsuccessful landing, 1 for a successful one
- Overall landing success rate across the cleaned dataset: 66.7%
- https://github.com/leonorrebola9/datascience_capstone/blob/main/labs-jupyter-spacex-Data%20wrangling-v2.ipynb

EDA with Data Visualization

- Scatter plots (Flight Number vs. Launch Site, Payload vs. Launch Site, Flight Number vs. Orbit Type, Payload vs. Orbit Type) were used to reveal how launch success (color-coded) relates to site, payload mass and orbit over time
- A bar chart of success rate by orbit type highlighted which orbits are most reliably associated with successful landings
- A line chart of yearly average success rate showed the overall improvement trend in landing success as SpaceX's technology matured
- https://github.com/leonorrebola9/datascience_capstone/blob/main/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

- Loaded the SpaceX.csv dataset into a SQLite database table (SPACEXTABLE) via `pandas.to_sql()`, then queried it with SQL magic (`%sql`)
- Queries covered: unique launch sites, sample records, total/average payload mass by customer and booster version, first successful ground-pad landing date, boosters with drone-ship success in a payload range, mission outcome counts, booster(s) with maximum payload, 2015 drone-ship failures by month, and a ranked count of landing outcomes over a date range
- https://github.com/leonorrebola9/datascience_capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- `folium.Circle` and `folium.Marker` were added for each of the 4 launch sites, using their latitude/longitude, to visualize their geographic spread and proximity to the equator/coast
- A `MarkerCluster` grouped individual launch records, color-coded green (success) or red (failure), so success patterns per site are visible at a glance
- `MousePosition`, a `PolyLine` and a distance marker were added to measure and display the distance from a launch site to its nearest coastline, highlighting SpaceX's preference for coastal/low-population launch sites
- https://github.com/leonorrebola9/datascience_capstone/blob/main/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

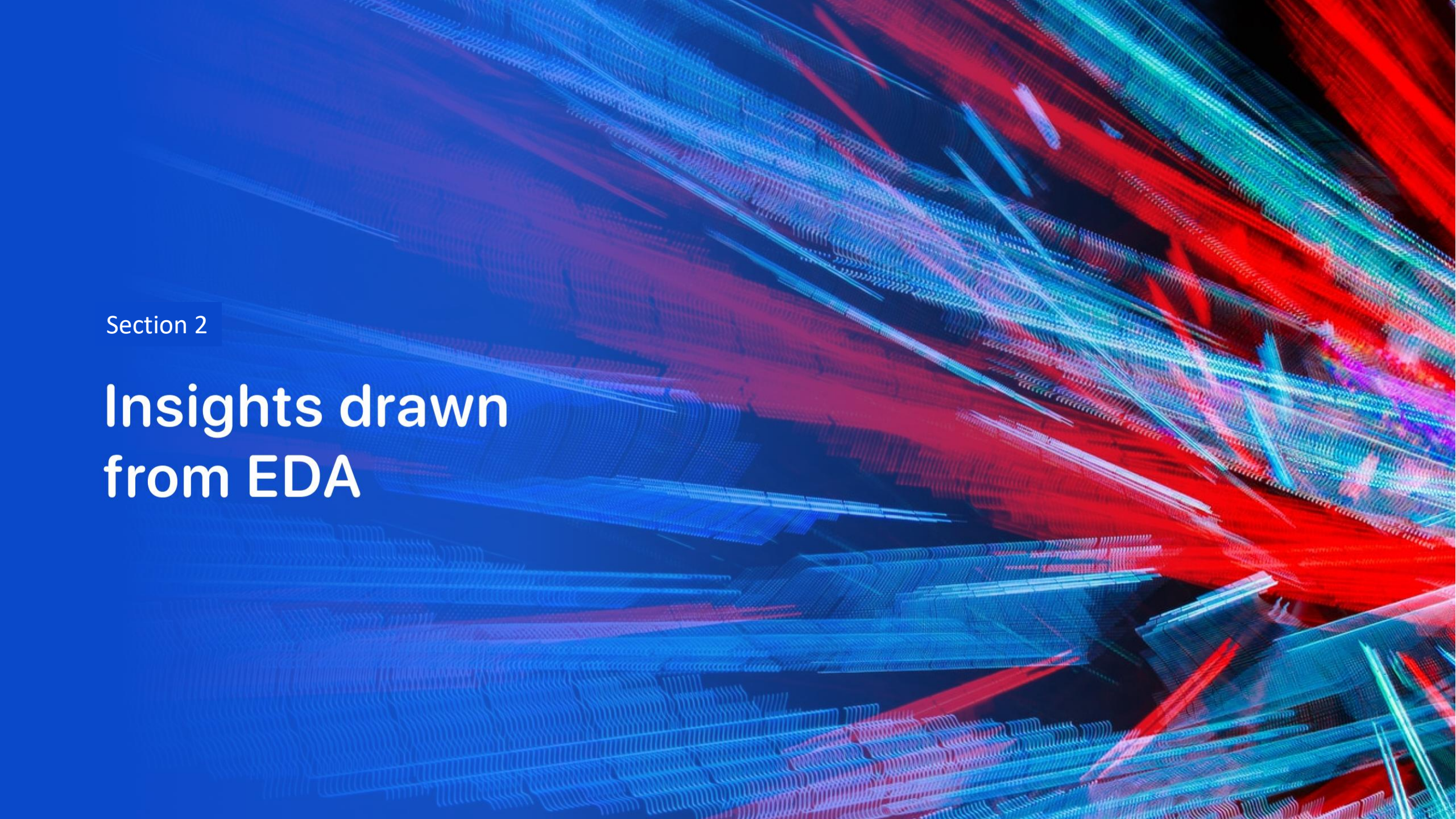
- A dropdown lets the user pick 'All Sites' or a single launch site, driving a pie chart of success counts (or success vs. failure split for one site)
- A range slider (0-10,000 kg) filters a scatter plot of Payload Mass vs. launch outcome (class), color-coded by Booster Version Category, updating interactively as the range changes
- These interactions let a user quickly compare success rates across sites and see how payload mass and booster version relate to landing success
- https://github.com/leonorrebola9/datascience_capstone/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

- Features were standardized with StandardScaler, then split into train/test sets (80/20, random_state=2)
- Four classifiers — Logistic Regression, SVM, Decision Tree, KNN — were each tuned via GridSearchCV (10-fold cross-validation) over a model-specific hyperparameter grid
- Each tuned model's cross-validated accuracy and test-set accuracy were compared, along with its confusion matrix, to select the best performer
- The Decision Tree Classifier had the highest cross-validated accuracy (87.5%), making it the recommended model
- https://github.com/leonorrebola9/datascience_capstone/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

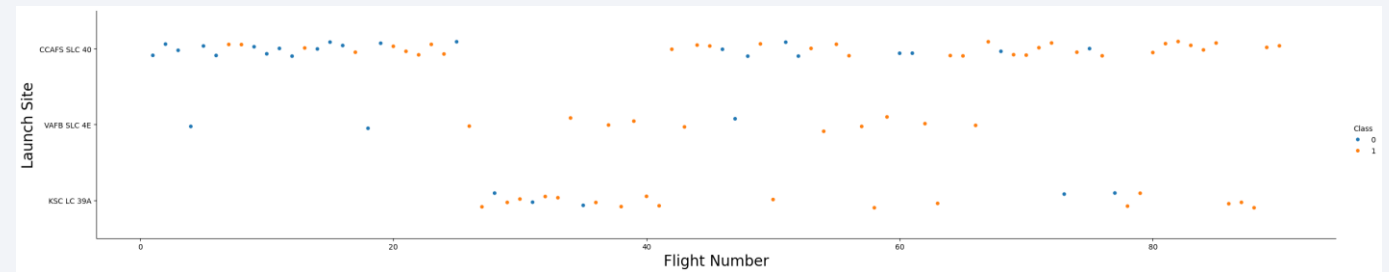


Section 2

Insights drawn from EDA

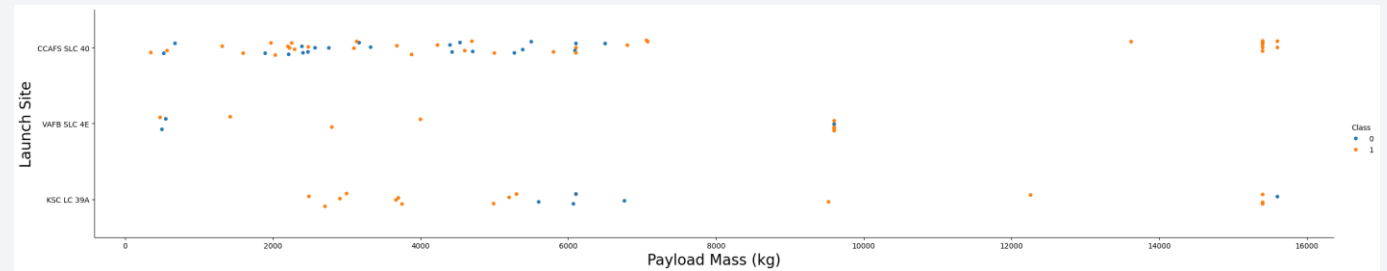
Flight Number vs. Launch Site

- Chart: seaborn catplot, Launch Site (y) vs Flight Number (x), colored by outcome (Class 0=blue=did not land, Class 1=orange=landed)
- CCAFS SLC-40 was used from flight #1 and shows a mix of outcomes throughout, with orange (success) becoming denser at higher flight numbers
- KSC LC-39A only came online around flight #28 onward — later in the program — and already shows a healthy share of successful (orange) landings



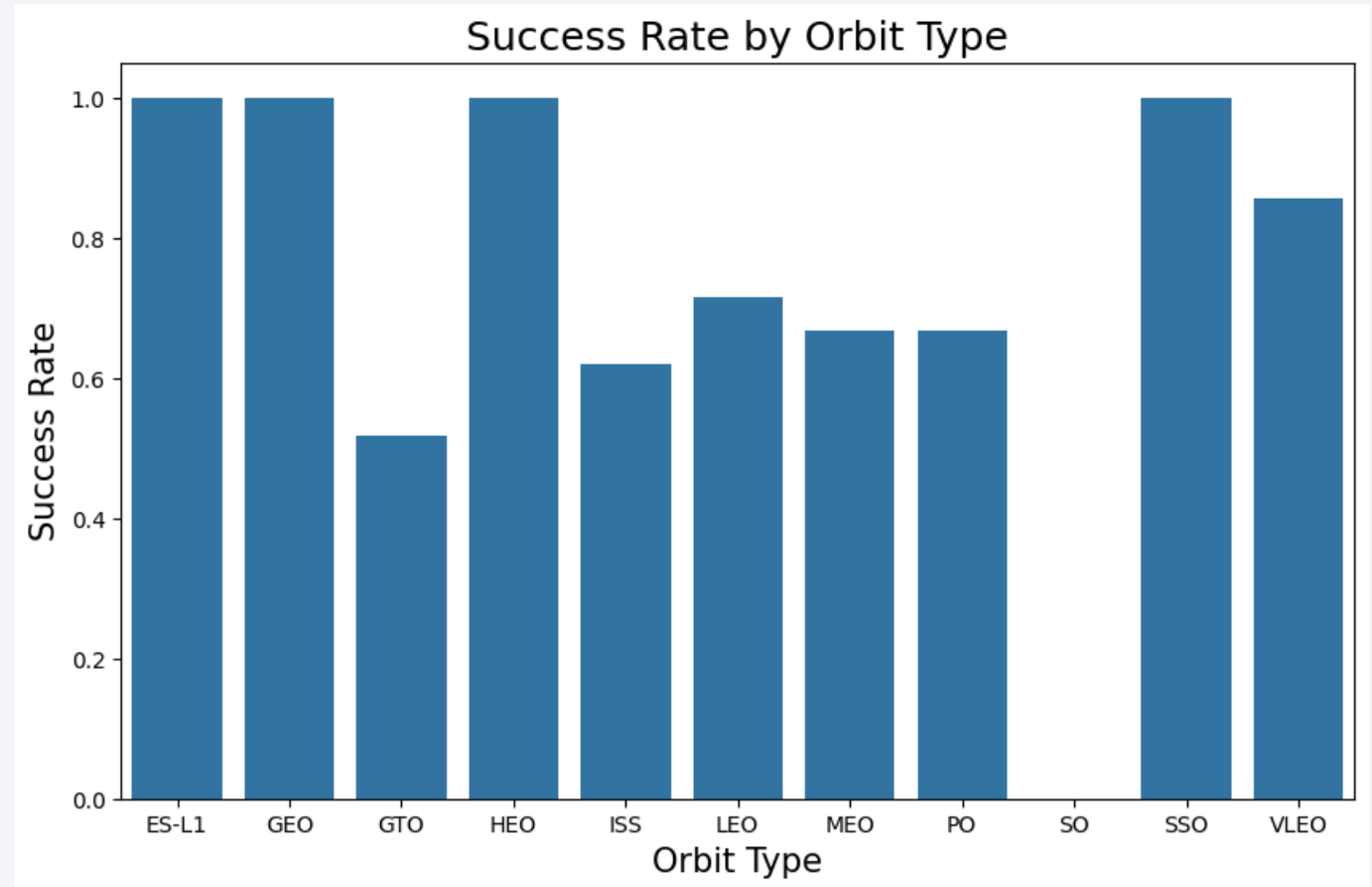
Payload vs. Launch Site

- Chart: seaborn catplot, Launch Site (y) vs Payload Mass (x), colored by landing outcome
- VAFB SLC-4E never launched payloads heavier than ~10,000 kg in this dataset
- CCAFS SLC-40 and KSC LC-39A both handle a wide range of payload masses, including the heaviest



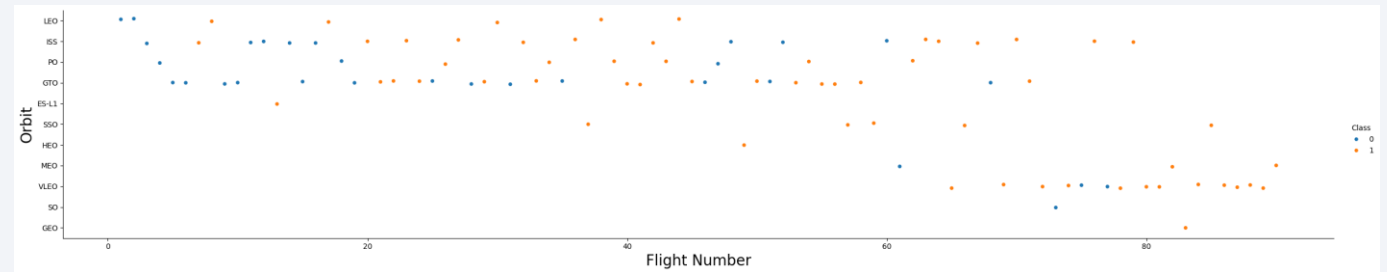
Success Rate vs. Orbit Type

- Chart: bar chart of mean Class (success rate) grouped by Orbit type
- ES-L1, GEO, HEO and SSO orbits show a 100% landing success rate in this dataset (though based on few launches)
- GTO has the lowest success rate (~52%); VLEO is strong (~86%) reflecting many recent, mature-era launches



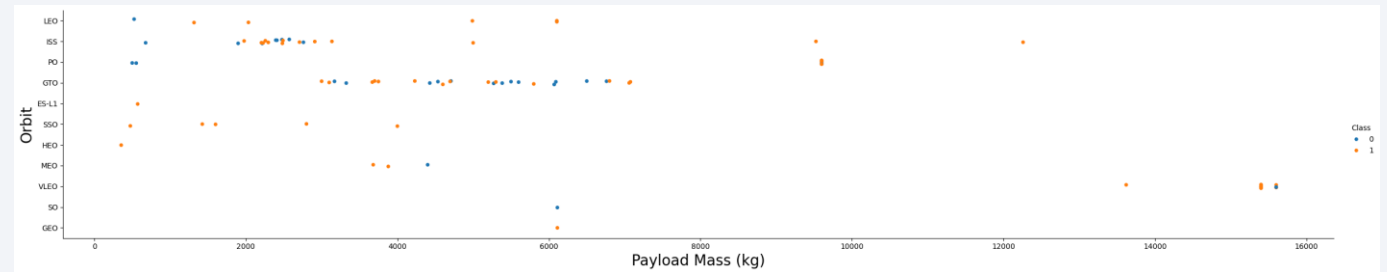
Flight Number vs. Orbit Type

- Chart: seaborn catplot, Orbit (y) vs Flight Number (x), colored by landing outcome
- In LEO, success is clearly related to flight number — later flights (higher numbers) succeed far more often
- GTO shows failures spread across both early and later flight numbers, a harder orbit to land from consistently



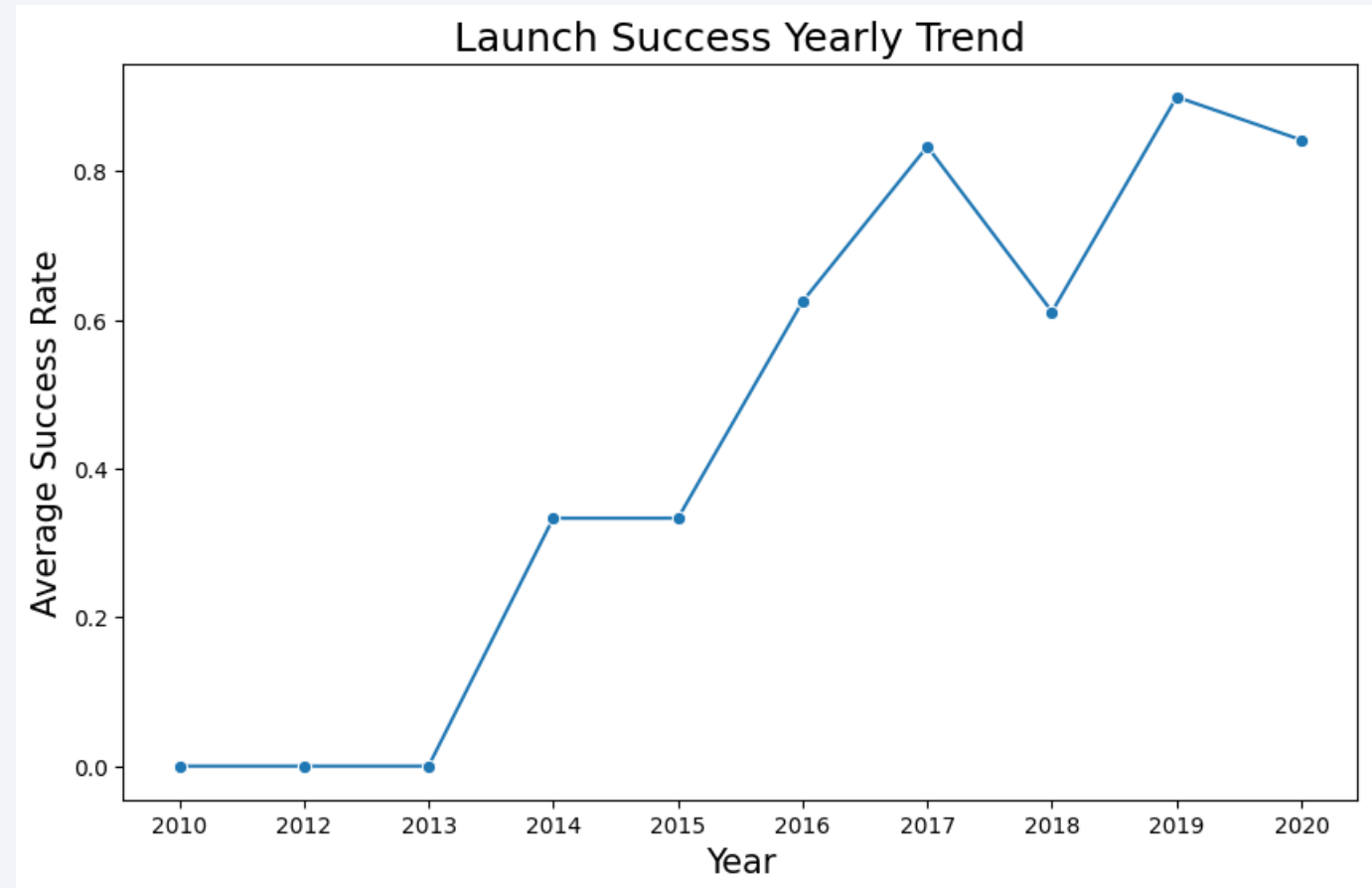
Payload vs. Orbit Type

- Chart: seaborn catplot, Orbit (y) vs Payload Mass (x), colored by landing outcome
- Heavier payloads to Polar, LEO and ISS orbits tend to have a higher positive landing rate
- It's hard to distinguish a clear payload-success pattern for GTO, given its mix of outcomes



Launch Success Yearly Trend

- Chart: line chart of average landing success rate by year
- Success rate rose sharply from 2013 to 2017, then stayed roughly stable through 2018-2020 (with a dip in 2018)
- This trend reflects SpaceX's iterative booster improvements (v1.0 → v1.1 → FT → Block 5)



All Launch Site Names

- Query: `SELECT DISTINCT Launch_Site FROM SPACEXTABLE;`
- Result: 4 unique launch sites — CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- These are the only pads SpaceX has used for Falcon 9/Heavy launches in this dataset

Launch Site Names Begin with 'CCA'

- Query: `SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`
- Result: 5 earliest CCAFS LC-40 launches (2010-2013), including the first-ever Falcon 9 flight
- Confirms CCAFS LC-40 was SpaceX's original launch pad before other sites were added

Total Payload Mass

- Query: `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer='NASA (CRS)';`
- Result: 45,596 kg total payload mass carried for NASA's Commercial Resupply Services (CRS) missions
- Highlights NASA CRS as one of SpaceX's major early customers

Average Payload Mass by F9 v1.1

- Query: `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version='F9 v1.1';`
- Result: 2,928.4 kg average payload mass for the F9 v1.1 booster version
- F9 v1.1 was an early-generation booster with more limited payload capacity than later versions

First Successful Ground Landing Date

- Query: `SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome='Success (ground pad)';`
- Result: 2015-12-22 — the date of SpaceX's first successful ground-pad landing
- This was a historic milestone, proving controlled first-stage recovery was viable

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query: boosters with Landing_Outcome='Success (drone ship)' AND payload mass between 4000-6000 kg
- Result: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- All four are 'Full Thrust' (FT) boosters, showing this generation reliably handled mid-range payloads on drone-ship landings

Total Number of Successful and Failure Mission Outcomes

- Query: `SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;`
- Result: Success — 98, Success (payload status unclear) — 1, Success (with extra note) — 1, Failure (in flight) — 1
- Mission (payload delivery) outcomes are overwhelmingly successful — 100/101 missions succeeded

Boosters Carried Maximum Payload

- Query: `booster_version` with the maximum `PAYLOAD_MASS__KG_` (via subquery with `MAX()`)
- Result: 12 booster serials tied for the maximum payload mass, all Block 5 (F9 B5) boosters, e.g. B1048.4, B1049.4, B1051.3, B1056.4...
- Shows Block 5 is SpaceX's highest-payload-capacity booster generation

2015 Launch Records

- Query: month, Landing_Outcome, Booster_Version, Launch_Site WHERE Landing_Outcome='Failure (drone ship)' AND year=2015
- Result: January — F9 v1.1 B1012 at CCAFS LC-40; April — F9 v1.1 B1015 at CCAFS LC-40
- Both 2015 drone-ship failures involved the earlier F9 v1.1 booster, before landing technology matured

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

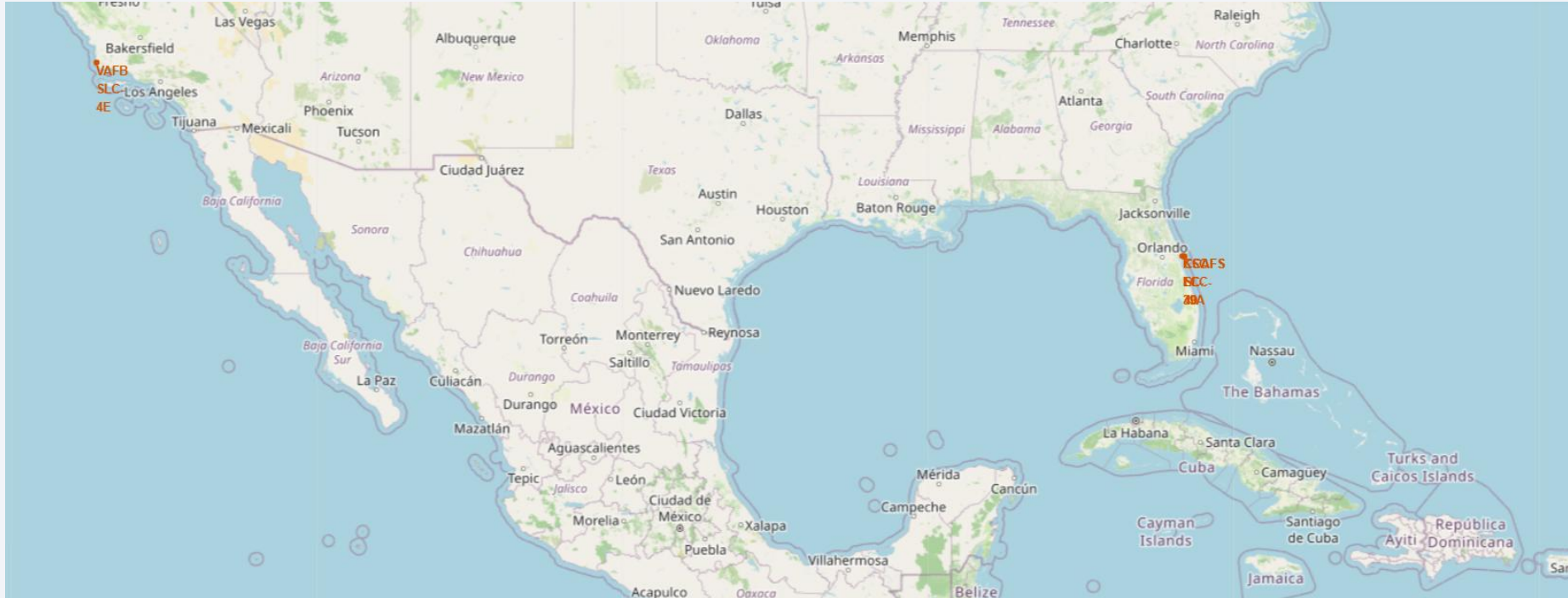
- Query: Landing_Outcome ranked by COUNT(*) between 2010-06-04 and 2017-03-20, descending
- Result (top to bottom): No attempt (10), Success (drone ship) (5), Failure (drone ship) (5), Success (ground pad) (3), Controlled (ocean) (3), Uncontrolled (ocean) (2), Failure (parachute) (2), Precluded (drone ship) (1)
- 'No attempt' still leads in this early period — landing attempts only became routine as the program matured

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear, dark blue sky.

Section 3

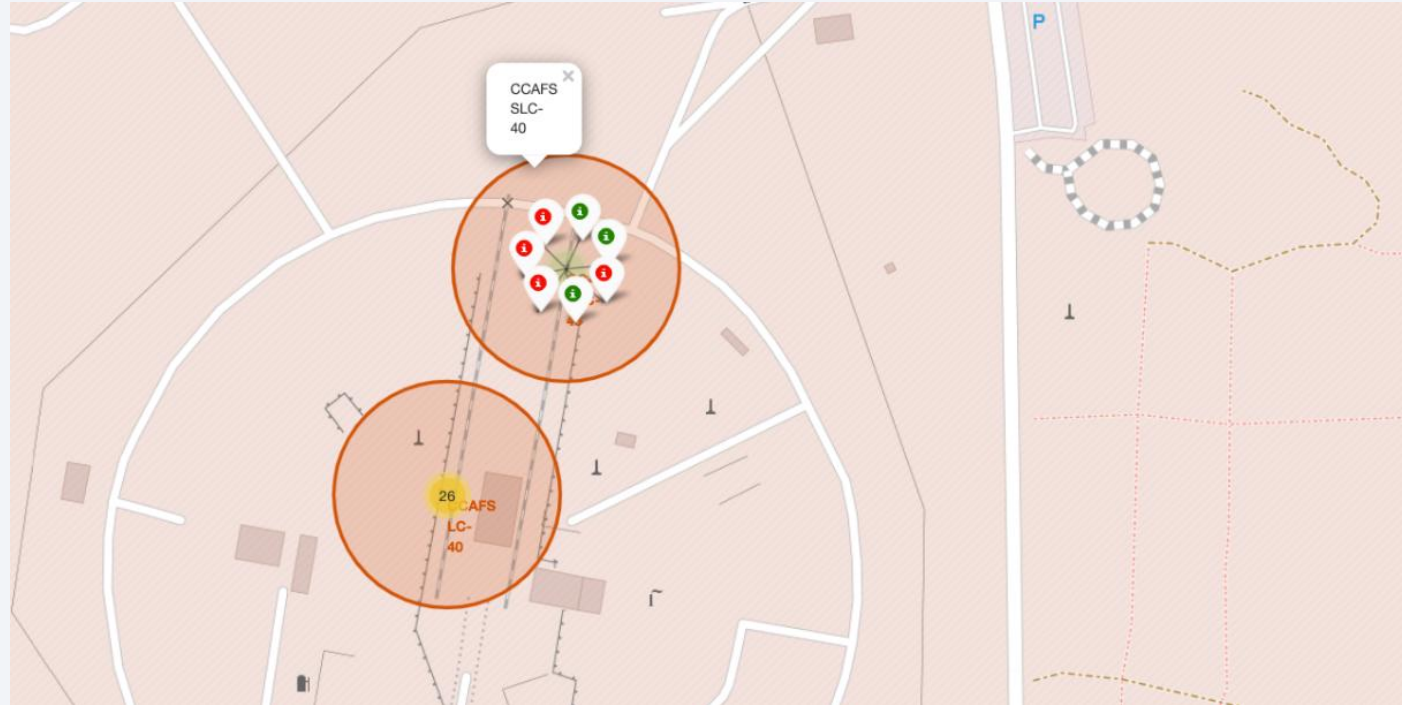
Launch Sites Proximities Analysis

All Launch Sites on a Global Map



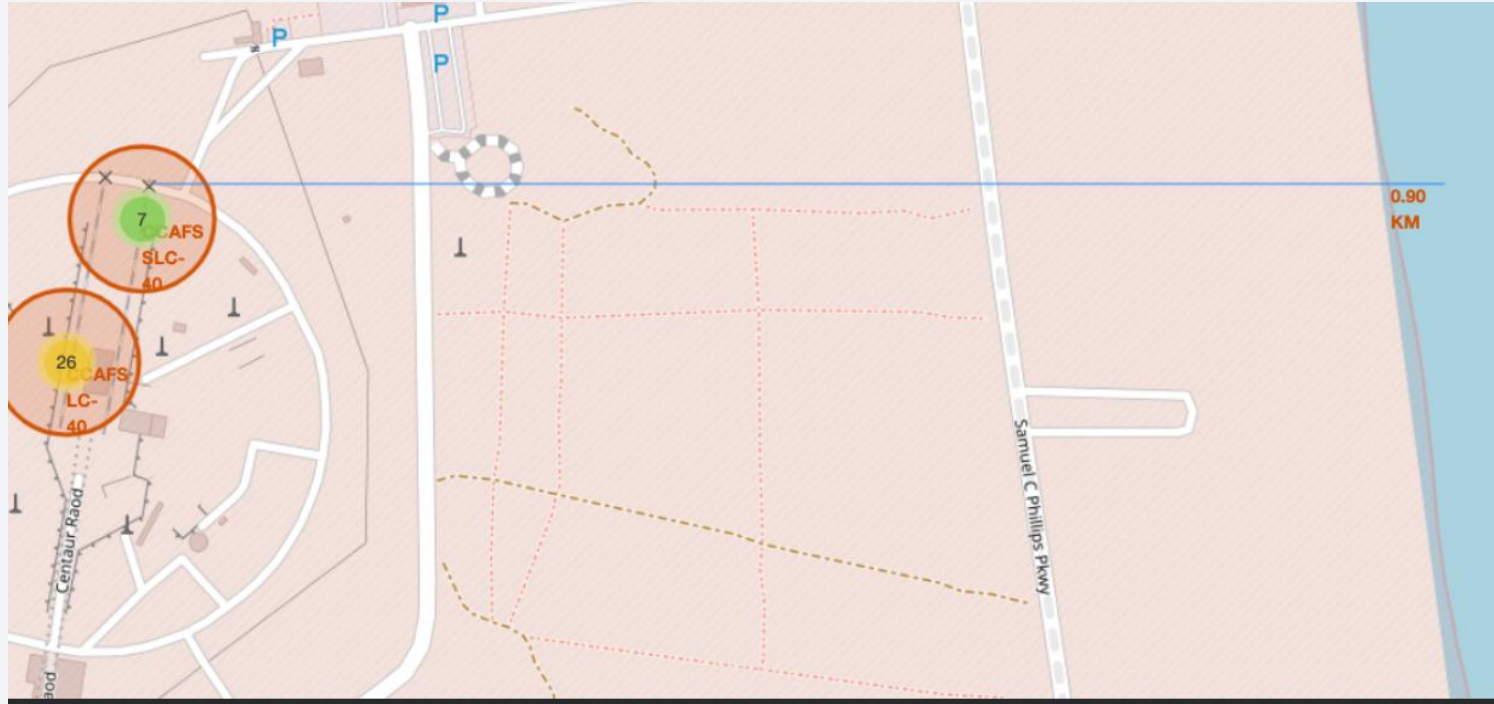
- Circle + text markers were added for all 4 Falcon 9 launch sites (CCAFS LC-40/SLC-40, KSC LC-39A, VAFB SLC-4E), overlaid on a map of the continental US
- All sites sit close to the coast (Florida east coast and California west coast), confirming SpaceX's preference for coastal, low-population launch corridors

Launch Outcomes by Site (Success/Failure Markers)

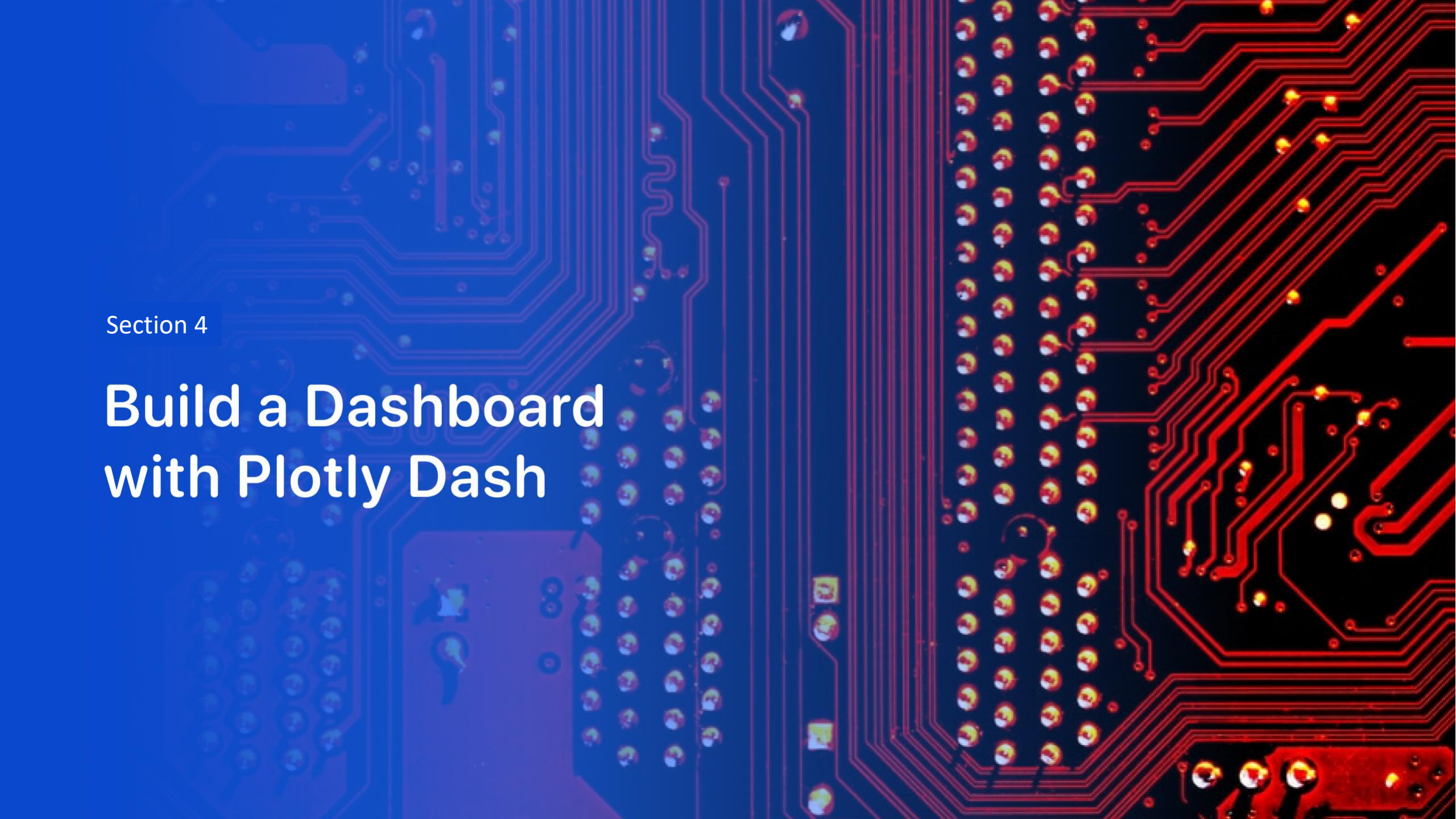


- Individual launch markers were color-coded green (success) and red (failure) and grouped with a MarkerCluster, so success patterns per site are visible when zoomed in
- Zooming into CCAFS shows a mix of green and red pins at SLC-40, illustrating both early failures and later successes at the same pad

Distance from Launch Site to Nearest Coastline



- MousePosition, a PolyLine and a distance marker were used to measure the distance from a launch pad to its nearest coastline point
- The measured distance here is 0.90 km, showing launch pads are sited extremely close to open water — consistent with range-safety practice for downrange rocket debris

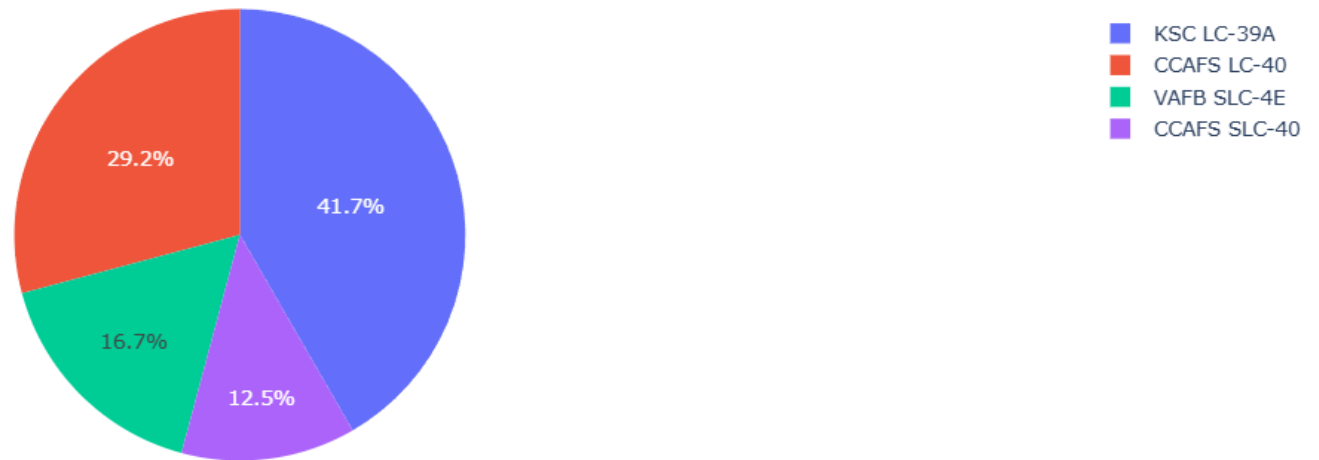


Section 4

Build a Dashboard with Plotly Dash

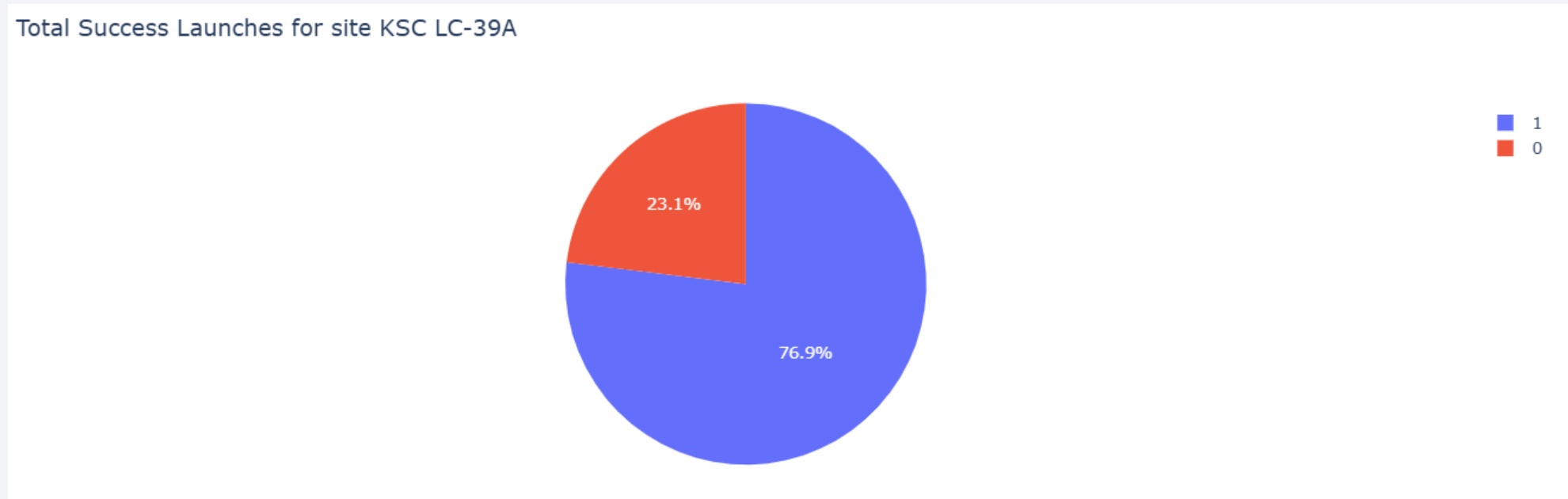
Launch Success Count — All Sites

Total Successful Launches by Site



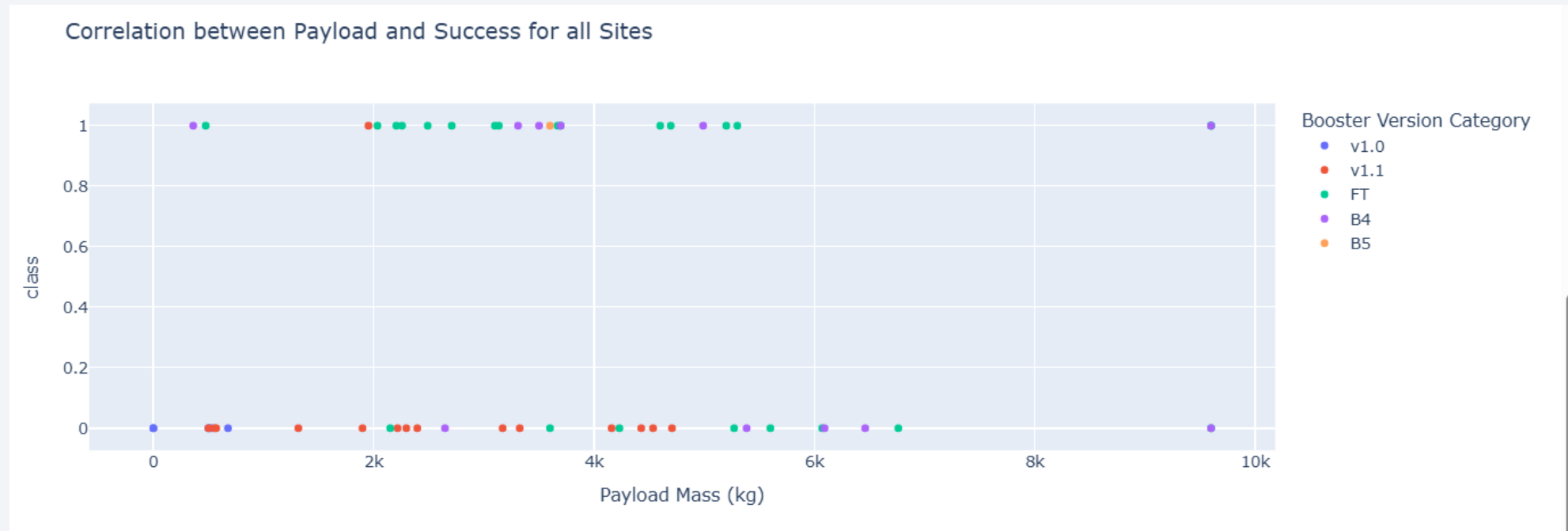
- With the dropdown set to 'All Sites', the pie chart shows the share of total successful launches contributed by each site
- KSC LC-39A contributes the most successful launches overall (41.7%), followed by CCAFS LC-40 (29.2%)

Highest Success-Rate Site — Pie Chart



- Selecting KSC LC-39A in the dropdown shows its individual success vs. failure split
- KSC LC-39A has the highest success rate of any site: 76.9% successful landings vs. 23.1% failures

Payload vs. Launch Outcome — Scatter Plot

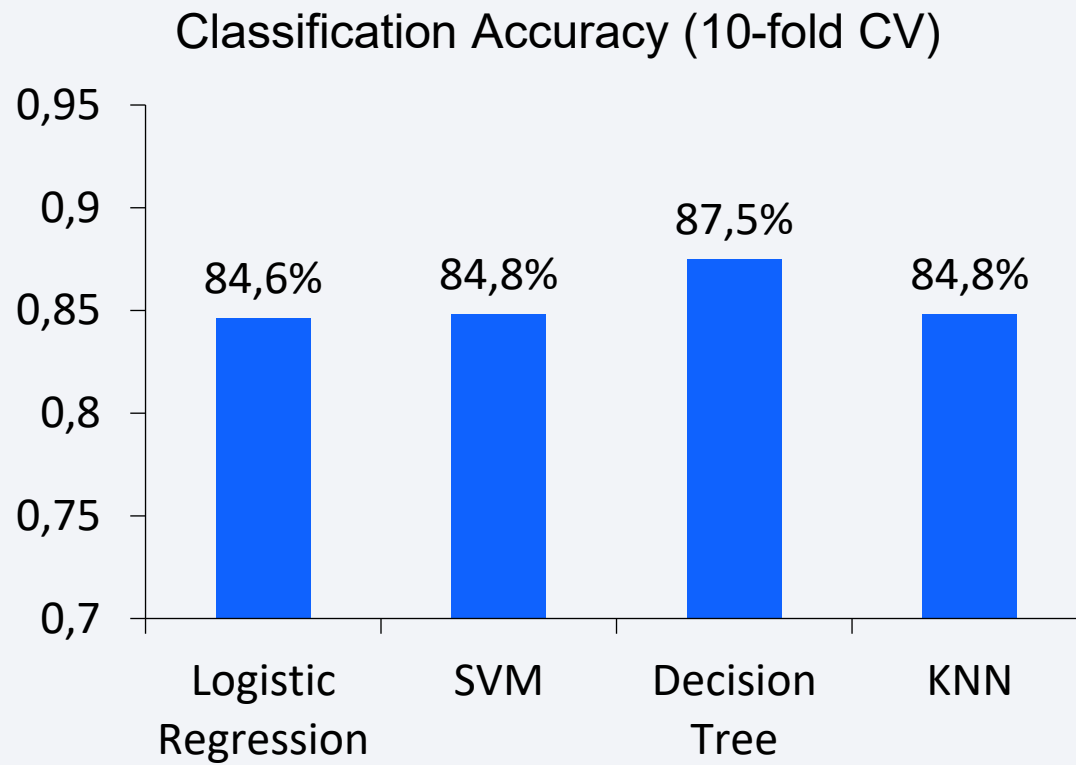


- The scatter plot (all sites, full payload range) plots landing outcome (class) against Payload Mass, colored by Booster Version Category
- Newer booster versions (FT, B4, B5) account for almost all successful (class=1) launches, while the earliest v1.0/v1.1 boosters cluster heavily among failures (class=0)

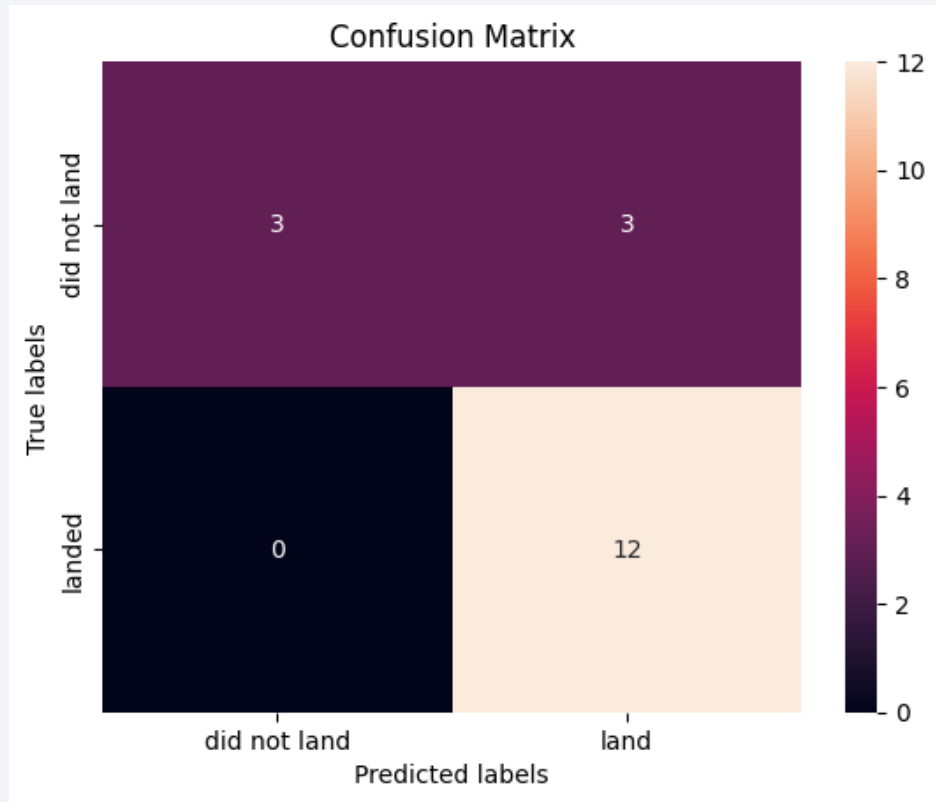
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



Best model: Decision Tree Classifier

Cross-validated accuracy: 87.5%

Test-set accuracy: 83.3% (tied with the other 3 models, due to the small 18-sample test set)

The confusion matrix shows the model correctly identifies most successful landings, with a small number of false positives/negatives typical of a modest dataset size

Conclusions

- Landing success rate has improved substantially over time as SpaceX refined its Falcon 9 booster (from earlier v1.0/v1.1 versions to the modern Block 5)
- Launch site, orbit type and payload mass all show a measurable relationship with landing outcome, with some orbits and payload ranges achieving much higher success rates than others
- SQL and visual EDA were consistent with each other, giving confidence in the cleaned dataset and the patterns identified
- The Decision Tree Classifier was the best-performing model (87.5% cross-validated accuracy), outperforming Logistic Regression, SVM and KNN, and is recommended for predicting Falcon 9 landing outcomes

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

